

Dayun Choi

✉ cdy3773@kaist.ac.kr | cdy1109@gmail.com

🌐 <https://choishio.github.io/>

🌐 <https://www.linkedin.com/in/dayun-choi-a9043b241/>

🐙 ChoiShio 📷 shio_choi



"I am a Master's–PhD integrated student in Electrical Engineering at KAIST, specializing in **deep learning for target sound extraction (TSE)** in complex acoustic scenes. My research focuses on utilizing **spatial queries**, such as direction-of-arrival (DoA), to extract desired multichannel signals, with potential applications in AR/VR. In addition to algorithmic development, I have hands-on experience in the deployment of speakers, sensors, and audio interfaces in **real-world experiments**, which has given me practical insight into audio signal processing. I am also interested in multimodal systems that integrate visual or textual cues, as well as generative methods for spatial audio, which I plan to explore further."

Education

Mar, 2022 – Present

📖 **Master's and PhD Integrated Course Student** in School of Electrical Engineering at Korea Advanced Institute of Science and Technology (KAIST)
Advisor: Prof. Jung-Woo Choi

Research Assistant at Smart Sound Lab

Focus: Deep Learning-based Target Sound Extraction on Multichannel Mixture using Spatial Queries

GPA: 3.88/4.3 (95.8/100) – via 93 credits.

Mar, 2018 – Feb, 2022

📖 **B.S. in Electronic Engineering** at Sogang University
Advisor: Prof. Yangmo Yoo

Undergraduate Intern at Medical Imaging Computing Systems (MICS) Lab

Research Project: Deep Learning-based Video Classification on Lung Ultrasound Imaging

GPA: 4.12/4.3 (98.6/100) – via 134 credits. (*Summa Cum Laude*)

Publications

Conference Proceedings

- 1 **D. Choi** and J.-W. Choi, "SoundCompass: Navigating target sound extraction with effective directional clue integration in complex acoustic scenes", in *Proc. International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, Barcelona, Spain, 2026, pp. 15 657–15 661. 📄 DOI: <https://doi.org/10.1109/ICASSP55912.2026.11461583>, 📺 Demo: <https://choishio.github.io/demo-SoundCompass/>.
- 2 **D. Choi** and J.-W. Choi, "Multichannel-to-multichannel target sound extraction using direction and timestamp clues", in *Proc. International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, Hyderabad, India, 2025, pp. 1–5. 📄 DOI: <https://doi.org/10.1109/ICASSP49660.2025.10890145>, 📺 Demo: https://choishio.github.io/demo_M2M-TSE/.
- 3 **D. Choi** and J.-W. Choi, "Target sound extraction on multichannel reverberant mixture", in *2024 Autumn Conference of the Acoustical Society of Korea (ASK)*, ASK, Busan, Korea, 2024, pp. 170–171.
- 4 **D. Choi** and J.-W. Choi, "Target sound extraction on reverberant mixture using sound-class label and timestamp information", in *2024 Spring Conference of the Acoustical Society of Korea (ASK)*, ASK, Jeju, Korea, 2024, pp. 335–336.

- 5 **D. Choi**, D. Lee, and J.-W. Choi, "Low complexity dnn model for multichannel speech enhancement", in *2023 Spring Conference of the Korean Society of Noise and Vibration Engineering (KSNVE)*, KSNVE, Samcheok, Korea, 2023, pp. 132–132.
- 6 D. Lee, **D. Choi**, and J.-W. Choi, "DeFT-AN RT: Real-time multichannel speech enhancement using dense frequency-time attentive network and non-overlapping synthesis window", in *Proc. Interspeech*, ISCA, Dublin, Ireland, 2023, pp. 864–868.  DOI: <https://doi.org/10.21437/Interspeech.2023-2437>.

Journal Articles

- 1 Y. Shul, **D. Choi**, and J.-W. Choi, "CST-former: Multidimensional attention-based transformer for sound event localization and detection in real scenes", *arXiv preprint arXiv:2504.12870*, 2025.  DOI: <https://doi.org/10.48550/arXiv.2504.12870>.
- 2 **D. Choi** and J.-W. Choi, "Target sound extraction on reverberant mixture", *The Journal of the Acoustical Society of America (JASA)*, vol. 154, no. 4_supplement, A270–A271, 2023.  DOI: <https://doi.org/10.1121/10.0023494>.

Research Projects

KAIST

- | | |
|-----------------------|--|
| Jul, 2024 – Present |  IoT-based Eavesdropping Attack Detection and Defense Technology funded by National Research Foundation of Korea (NRF) and STEAM research grant with National University of Singapore (NUS) <ul style="list-style-type: none"> Developed DNN models for speech dialogue restoration and enhancement from adjacent indoor spaces Generated simulated and real-world datasets using Pyroomacoustics library, loudspeakers, and multiple IoT sensors to evaluate system viability in real-world environments Developed software program user interface and hardware products Tools: Python (PyTorch), Matlab, NI SignalExpress, LabVIEW KRK ROKIT 5 (RP5 G3), RME Audio FIREFACE UC, NI 9234, Sensors (352C66, 355B04, HY929) |
| May, 2024 – Present |  General-Purpose AI for Integrated Analysis and Generation of Spatial Audio funded by National Research Foundation of Korea (NRF) <ul style="list-style-type: none"> Developed Universal Spatial Audio Analysis Network to infer source directions, positions, classes, and room acoustics, and separate signals into object-oriented formats Developed Universal Spatial Audio Generation Network to synthesize realistic 3D audio from single-channel or separated sources, supporting multi-modal input and 6-DoF spatial rendering Tools: Python (PyTorch) |
| Jan, 2023 – Apr, 2026 |  Noise Cancellation and Speech Recognition Technology Robust to Battlefield Environments funded by Center for Applied Research in Artificial Intelligence (CARAI) <ul style="list-style-type: none"> Developed DNN models for robust noise cancellation in battlefield scenarios Tools: Python (PyTorch) |

Research Projects (continued)

Sogang University

- Sep, 2021 – Dec, 2021  **Real-time COVID-19 Detection System from Lung Ultrasound Videos**
- Developed DNN models for video classification of lung ultrasound data
 - Explored model compression techniques for real-time deployment
 - Developed system user interface
 - Tools: Python (TensorFlow, PyQt)
- Jul, 2021  **AI-based Automobile Recognition System** in The 2nd Software Pre-Education Program for Hyundai Mobis - Sogang University (FoSS)
- Pre-trained DNN models for object detection using Open-Images dataset
 - Deployed model on Jetson Nano and validated automobile detection performance
 - Tools: Python (PyTorch), Jetson Nano

Assistantships

KAIST

- Summer, 2025  [EE495] Individual Study (*Advisor: Prof. Jung-Woo Choi*)
- Spring – Winter, 2025  **Lab Head** of Smart Sound Lab (*Advisor: Prof. Jung-Woo Choi*)
- Winter, 2024  [EE495] Individual Study (*Advisor: Prof. Jung-Woo Choi*)
- Autumn, 2024  [EE202] Signals and Systems (*Lecturer: Prof. Jung-Woo Choi*) (**Head TA**)
- Spring, 2024  [EE488] Introduction to Audio Signal Processing (*Lecturer: Prof. Jung-Woo Choi*) (**Head TA**) & [EE495] Individual Study (*Advisor: Prof. Jung-Woo Choi*)
- Winter, 2023  [EE202] Signals and Systems (*Lecturer: Prof. Jung-Woo Choi*) (**Head TA**)
- Autumn, 2023  [EE202] Signals and Systems (*Lecturer: Prof. Jung-Woo Choi*)
- Spring, 2023  [EE305] Introduction to Electronics Design Lab (*Lecturer: Prof. Jun-Bo Yoon*)
- Autumn, 2022  [EE202] Signals and Systems (*Lecturer: Prof. Jung-Woo Choi*)

Honors & Awards

KAIST

- Mar 2nd, 2024  **The 1st place** at Technical Committee on Architectural Acoustics Student Paper Award for December Sydney meeting

Sogang University

- Feb 21st, 2022  Honors
- Mar 30th, 2020  **Dean's List** for Electronic Engineering
- Mar, 2020 – Feb, 2022  National Scholarship for Science and Engineering
- Mar 29th, 2019  **Dean's List** for Electronic Engineering
- Nov 29th, 2018  **Dean's List** for Electronic Engineering
- Nov 16th, 2018  **The 3rd Prize** at the 7th Proposal Contest of General Elective Subjects

Miscellaneous Experiences

KAIST

- Jun 18th, 2024  **Representative Presenter (Oral)** of Smart Sound Lab for KAIST-NUS Joint Seminar
- Apr 5th, 2024  **Representative Presenter (Oral)** of Smart Sound Lab for the 5th SORI Workshop
- Jul 13th, 2023  **Representative Presenter (Poster)** of Smart Sound Lab for the 4th SORI Workshop

Sogang University

- Jul 19th – 30th, 2021  The 2nd Software Pre-Education Program for Hyundai Mobis - Sogang University (FoSS)
- Jun 29th – Jul 9th, 2019  The 24th Overseas Culture Tour to Greece & Türkiye
- Jan, 2019 – Feb, 2020  **Head of education and culture department** of Hyunwoohoe (Classical Guitar Club)

Skills

- Programming Languages  Python/Matlab (Advanced), C/VHDL/Linux/HTML (Intermediate)
- ML Frameworks  PyTorch/TensorFlow (Advanced), PyTorch Lightning (Intermediate)
- Software  OrCAD/PSpice (Advanced), NI SignalExpress/Audacity (Intermediate), nanoHUB/Unity/LabVIEW (Elementary)
- Hardware  Loudspeaker: KRK ROKIT 5 (RP5 G3)/Audio Interface: RME Audio FIREFACE UC/NI Data Acquisition Tool: NI 9234/Sensor: 352C66, 355B04, HY929, Polytec PDV-100/Microphone Array: ReSpeaker Mic Array v2.0 (Intermediate), Arduino/EasyFPGA-Combo2/Jetson Nano (Elementary)
- Languages  Korean (Native), English (TOEIC: 735 – expired)